***Problem:*** Consider a homogeneous, isotropic thin plate of thickness $t$ subjected to a transverse distributed load $p(x, y)$. The coordinate $z$ is measured through the thickness of the plate with the origin located at the mid-plane, such that $-t/2 \leq z \leq t/2$. Assume that the Kirchhoff assumptions of classical plate theory are valid. The three-dimensional equilibrium equations in the absence of body forces are given by

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} = 0, \tag{1a}$$

$$\frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yz}}{\partial z} = 0, \tag{1b}$$

$$\frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} = 0. \tag{1c}$$

The in-plane bending stresses $\sigma_x$, $\sigma_y$, and $\tau_{xy}$ are assumed to be known functions of $x$, $y$, and $z$.

(a) Using the above equilibrium equations together with the boundary conditions on the top and bottom surfaces of the plate ($z = \pm t/2$), derive the expressions for the transverse shear stresses $\tau_{xz}$, $\tau_{yz}$, and the normal stress $\sigma_z$ through the plate thickness.

(b) By integrating the transverse shear stresses through the plate thickness, define the resultant shear forces

$$Q_x = \int_{-t/2}^{t/2} \tau_{xz} \, dz, \tag{2a}$$

$$Q_y = \int_{-t/2}^{t/2} \tau_{yz} \, dz, \tag{2b}$$

and use the transverse equilibrium of the plate to obtain the governing differential equation of plate bending in terms of the transverse deflection $w(x, y)$.

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**Solution:**
a) From previous calculations, we have

$$\sigma_x = -\frac{Ez}{1-\nu^2} \left( \frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right), \tag{3a}$$

$$\sigma_y = -\frac{Ez}{1-\nu^2} \left( \frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right), \tag{3b}$$

$$\tau_{xy} = -\frac{Ez}{1+\nu} \frac{\partial^2 w}{\partial x \, \partial y}. \tag{3c}$$

By substituting Eqs. (3a) and (3c) into Eq. (1a), $\tau_{xz}$ can be obtained

$$\boxed{\tau_{xz} = -\frac{E}{2(1-\nu^2)}\left(\frac{t^2}{4} - z^2\right)\left[\frac{\partial}{\partial x}\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\right)\right]} \tag{4}$$

We know

$$Q_x = -D\frac{\partial}{\partial x}(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}) \rightarrow \frac{Q_x}{D} = -\frac{\partial}{\partial x}(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}),\ D = \frac{Et^3}{12(1-\nu^2)}$$

Then,

$$\boxed{\tau_{xz} = \frac{3Q_x}{2t}\left[1 - \left(\frac{2z}{t}\right)^2\right]} \tag{5}$$

By substituting Eqs. (3b) and (3c) into Eq. (1b), $\tau_{yz}$ can be obtained

$$\boxed{\tau_{yz} = -\frac{E}{2(1-\nu^2)}\left(\frac{t^2}{4} - z^2\right)\left[\frac{\partial}{\partial y}\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\right)\right]} \tag{6}$$

We know

$$Q_y = -D\frac{\partial}{\partial y}(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}) \to \frac{Q_y}{D} = -\frac{\partial}{\partial y}(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}),\ D = \frac{Et^3}{12(1-\nu^2)}$$

Then,

$$\boxed{\tau_{yz} = \frac{3Q_y}{2t}\left[1 - \left(\frac{2z}{t}\right)^2\right]} \tag{7}$$

Now, by substituting Eqs. (4) and (6) into Eq. (1c), $\sigma_z$ can be obtained

$$\Rightarrow$$

$$\boxed{\sigma_z = -\frac{E}{2(1-\nu^2)}\left(\frac{t^3}{12} - \frac{t^2 z}{4} + \frac{z^3}{3}\right)\left[\Big(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\Big)\Big(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\Big)\right]} \tag{8}$$

We know

$$\Big(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\Big)\Big(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\Big) = \nabla^4 w,\ \nabla^4 w = \frac{p}{D},\ D = \frac{Et^3}{12(1-\nu^2)}$$

Then,

$$\boxed{\sigma_z = -\frac{3p}{4}\left[\frac{2}{3} - \frac{2z}{t} + \frac{1}{3}\left(\frac{2z}{t}\right)^3\right]} \tag{9}$$

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b) Substituting Eq. (4) into Eq. (2a),

$$\boxed{Q_x = -\frac{Et^3}{12(1-\nu^2)}\frac{\partial}{\partial x}(\nabla^2 w) = -D\frac{\partial}{\partial x}(\nabla^2 w)} \tag{10}$$

And by substituting Eq. (6) into Eq. (2b),

$$\boxed{Q_y = -\frac{Et^3}{12(1-\nu^2)}\frac{\partial}{\partial y}(\nabla^2 w) = -D\frac{\partial}{\partial y}(\nabla^2 w)} \tag{11}$$

Finally, by writing BC at $z = -t/2$ in Eq. (8),

$$\Rightarrow$$

$$\boxed{\nabla^4 w = \frac{p}{D}} \tag{12}$$